

OS PYQ Traps and System Utilities — Detailed Lecture Notes

JKSSB Computer Awareness

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52.1 Why this lesson is a trap drill

The computer section of the JKSSB papers does not only test whether a candidate knows a term. It tests whether the candidate can tell a term apart from its *nearest neighbour*. A typical wrong option is not a random word; it is the close confusion that “sounds right” until the exact wording is checked. Device driver versus hardware, DMA versus ordinary CPU copying, system software versus application software, and the Windows Run command `winword` are all examples of this pattern.

Because of this, the present lesson is a **trap drill**. Every question in the practice set is built around a real near-neighbour or defective-option pattern catalogued in the course knowledge base. The aim is not to learn a large amount of new material, but to build the habit of reading each option as a *claim to be tested*. The rest of these notes explain the small set of facts the traps are built on, one distinction at a time.

Definition (Trap drill). *A study method in which each practice option is treated as a factual claim, and the candidate decides whether the claim is exactly correct, closely correct but wrong, or plainly wrong.*

52.2 The trap catalogue for this lesson

The traps this lesson drills on are collected below. Each row states the tempting wrong belief, the correct distinction that defeats it, and the knowledge-base reference where the pattern is recorded. Read the table as a checklist: if an option matches a row on the left, it is the trap, and the right-hand column tells you what to say instead.

Tempting wrong belief	Correct distinction	Ref.
A device driver is hardware.	A device driver is software (system software); the device is hardware.	TRAP-10
DMA makes the CPU copy data “faster”.	DMA moves data between an I/O device and main memory, bypassing the CPU.	§52.5
The OS runs from secondary storage.	While running, the OS resides in primary memory (RAM).	PYQ-22
Utility software is application software.	Utility software maintains the system; application software does the user’s task.	§52.3
MS Word is system software.	MS Word is application software.	§52.3
MS Access is an operating system.	MS Access is a database application, not an OS.	§52.7
Windows or macOS is a Linux distribution.	Ubuntu, Fedora, and Debian are Linux distributions.	§52.7
winword opens WordPad or Notepad.	winword opens MS Word.	§52.9
A component is internal or external by importance.	The classification follows position relative to the system unit.	§52.8
An RTOS is chosen for office or graphics work.	An RTOS is chosen for fixed time deadlines.	§52.6

52.3 The operating system: role and classification

The **operating system (OS)** is the master **system software** of a computer. It sits between the user (and the application programs) on one side and the hardware on the other. Its two central jobs are to provide a *user-friendly interface* and to manage the machine’s resources.

The interface may be a graphical one, such as the Windows desktop, or a command line, such as the MS-DOS prompt or the Linux shell. Either way, the user does not need to drive the hardware directly. In its resource-management role, the OS schedules the processor, allocates main memory, organises storage, and controls input and output devices from a single set of instructions. It also provides services — opening a file, printing a document, sending data over a network — that application programs call instead of talking to the hardware themselves.

Definition (Operating system). *The system software that controls the computer’s hardware and provides a user-friendly interface and a set of services to application programs.*

Common operating systems are Windows, Linux, UNIX, MS-DOS, Android, and macOS. An application program is not an operating system. MS Word, MS Excel, MS Access, a web browser, and a media player are applications, not operating systems. This is one of the exact-recognition distinctions the exam uses.

A frequently tested additional fact is *where* the OS is when the computer is running. The OS is stored permanently on secondary storage (the disk), but while the machine is running it is loaded into **primary memory (RAM)**. The CPU executes instructions only from main memory, so the running OS must be resident there. When power is removed, the RAM copy is lost and the OS is loaded again from disk at the next boot.

52.4 System, application, and utility software

Software is grouped by its purpose. **System software** runs and manages the machine: the operating system, device drivers, utility programs, and language translators (compiler, interpreter, assembler) all belong here. **Application software** performs a task the user wants done: typing a document, preparing a spreadsheet, browsing the web. **Utility software** is a small maintenance category: tools such as Disk Cleanup, Disk Defragmenter, and antivirus programs that keep the system healthy and efficient. Utility software is closely tied to system software, but it is useful to name it separately because exam options mix the three categories.

Category	Role	Examples
System software	Runs and manages the machine	OS, device drivers, translators
Application software	Does a user's task	MS Word, Excel, a browser
Utility software	Maintains and optimises the system	Disk Cleanup, Defragmenter, antivirus

Example 52.4.1. *An item asks, “Which of the following is application software?” with the options Windows, MS Word, Disk Defragmenter, and a device driver. Windows is an operating system (system software); Disk Defragmenter is utility software; a device driver is system software too. Only MS Word is application software. The trap is that all four options are well-known pieces of software, and only the category separates them.*

52.5 Device drivers are software

A **device driver** is a small software program that allows the operating system to communicate with a specific hardware device. Each device — a printer, a keyboard, a graphics card, a network adapter — needs a matching driver. The driver receives general commands from the OS and translates them into the device-specific instructions that the hardware understands. Without the correct driver, the OS cannot use the device properly.

The trap here is a classification trap. A device driver *controls* hardware, but the driver itself is **software**, and because it extends the operating system's control of the machine it is counted as **system software**. The device it drives is hardware. The OS itself is system software. An option that says “device driver is hardware” has confused the software with the device.

Definition (Device driver). *A software program that lets the operating system control a particular hardware device by translating OS commands into device-specific instructions.*

52.6 DMA: transferring data without the CPU

Normally, moving data between an input/output device and main memory keeps the CPU busy: the CPU reads each unit from the device and writes it to memory (or the reverse), one step at a time. **DMA** — **Direct Memory Access** — replaces this with a dedicated transfer. A DMA controller moves data *directly* between the I/O device and main memory while the CPU is bypassed for the duration of the transfer. The CPU starts the transfer and is notified by an interrupt when it is complete.

Definition (Direct Memory Access (DMA)). *A method of data transfer in which a controller moves data between an input/output device and main memory directly, without the CPU handling each individual byte.*

The advantages are clear. Because the CPU is not occupied with copying each byte, it is free to run other instructions, and large block transfers finish faster. Several traps grow from this description. DMA moves data between a *device and main memory*, not between the CPU and the device. The CPU is *bypassed* — it is not made “faster”. DMA does not increase the size of RAM and does not encrypt data. The CPU still initiates the transfer and handles the completion interrupt; it is simply not in the per-byte path.

52.7 Real-time operating systems

A **real-time operating system (RTOS)** is an operating system designed so that tasks complete within **fixed time deadlines**. In an ordinary desktop OS, a task may take as long as the scheduler permits as long as it finishes eventually. In an RTOS, meeting the deadline is the whole point: a late response is a failure, even if the answer is correct.

An RTOS is therefore chosen for embedded and control systems, such as industrial machinery, medical devices, and vehicle electronics, where a delayed action can be dangerous. It is not chosen for typing a document, designing a website, or editing a photograph; those are ordinary application tasks that do not carry hard deadlines.

Definition (Real-Time Operating System (RTOS)). *An operating system that guarantees that critical tasks complete within a bounded, fixed time so that deadlines are always met.*

52.8 DOS, UNIX, and Linux distributions

DOS stands for **Disk Operating System**. MS-DOS was a command-line operating system: the user typed commands rather than clicking icons. **UNIX** is a multi-user, multitasking operating system that influenced many later systems. **Linux** is an open-source, UNIX-like operating system; because its source can be freely used and modified, it is distributed in many packaged forms called **distributions**. Well-known Linux distributions include **Ubuntu**, **Fedora**, **Debian**, and **Linux Mint**.

This gives another exact-recognition drill. Windows and macOS are operating systems, but they are *not* Linux distributions. MS-DOS is an operating system but, again, not a Linux distribution. Only Ubuntu, Fedora, Debian, and Mint belong on the Linux-distribution list. Similarly, MS Access is a database application and is **not** an operating system at all, even though it is a familiar Microsoft product.

52.9 Internal and external hardware

Hardware is classified by its *position* relative to the system unit (the cabinet that houses the main machine). **Internal** hardware sits inside the system unit: the motherboard, the CPU, RAM, the hard disk, the power supply, and expansion cards. **External** hardware, also called a **peripheral**, sits outside the system unit and is joined to it through ports and cables: the monitor, keyboard, mouse, printer, and scanner.

The trap is that a familiar word can appear in the wrong column. A monitor and a keyboard are external; a motherboard and a RAM module are internal. The classification follows the position, not the importance of the component.

Definition (Internal vs external hardware). *Internal hardware is located inside the system unit; external hardware (a peripheral) is located outside it and connected through ports.*

52.10 The Windows Run box and the winword command

The Windows **Run** box, opened with the Windows + R shortcut, starts a program when the user types that program's command. The command **winword** opens **MS Word**. In the same way, **notepad** opens Notepad, **wordpad** opens WordPad, and **cmd** opens the Command Prompt. The command must match the program exactly.

Example 52.10.1. *An item asks which program is opened by typing **winword** in the Run box, offering Notepad, MS Word, WordPad, and File Explorer. The answer is **MS Word**. Notepad is opened by **notepad**, WordPad by **wordpad**, and File Explorer by **explorer**. The near-neighbours are deliberately word-processor or editor names, so the candidate who only half-remembers the command can be pulled to the wrong option.*

52.11 Running example: the office desktop

It helps to hang these facts on one concrete machine. Picture the office desktop used throughout this course: a multi-core PC running Windows, with MS Word and MS Excel installed, a printer, and an internet connection.

When the machine is switched on, the Windows OS is loaded from the disk into primary memory and stays resident there while the machine runs. MS Word and MS Excel are *application* software that the user opens to do work. Disk Cleanup and Disk Defragmenter are *utility* software that the user runs occasionally to keep the machine healthy. The printer does not work until its *device driver* — a piece of system software — is installed, so that Windows can translate its print commands into the printer's own instructions. When a large print job or file copy is under way, a *DMA* transfer can move the data between the device and main memory while the CPU continues with other work.

Example 52.11.1. *A single scenario can therefore generate several trap options about one machine. “Word is system software” is false — Word is application software. “The printer driver is hardware” is false — the driver is software. “The OS is stored on the disk and run from there” is half-true and therefore defective: it is stored on the disk but run from main memory. Recognising which category each item belongs to settles all three.*

52.12 Defective options and how to read them

Some official options are not merely wrong; they are *defective* — the wording itself is inconsistent or ambiguous. The knowledge base records this as a real feature of the papers (Anti-Pattern AP-02). A defective option is teaching material, not a fact to memorise. The safe method is to quote the option exactly, mark the defect, and then apply the stable concept.

Example 52.12.1. *A defective option might read, “The device driver is the hardware that stores the instructions the operating system uses to control a device.” The defect is the phrase “the hardware that stores the instructions”. A driver is software that holds those instructions; it is not hardware. The stable concept to apply is TRAP-10: a device driver is software. Once the defect is spotted, the option can be rejected without any doubt.*

The same discipline applies to near-neighbour options that are not defective but simply close. When two options are both plausible, compare them word by word against the definitions in this lesson. The distinction is usually a single term — *software* versus *hardware*, *device* to *memory* versus *CPU* to *device*, *application* versus *utility*, or **winword** versus **wordpad**. Reading for that one term is what turns a guess into a correct answer under negative marking.

52.13 Terminology ladder for this lesson

Direct-recall expansion items are the single most common JKSSB format, so the exact terms from this lesson are collected here with their full forms. Each row is one fact.

Term	Full form	One-line meaning
OS	Operating System	System software that provides an interface and manages resources
DMA	Direct Memory Access	Data moves between an I/O device and main memory with the CPU bypassed
RTOS	Real-Time Operating System	An OS that meets fixed time deadlines
DOS	Disk Operating System	A command-line operating system (MS-DOS)
Driver	Device driver	Software that lets the OS control a specific device
Utility	Utility software	Maintenance software such as Disk Cleanup or Defragmenter
Peripheral	External hardware	A device outside the system unit, joined by ports

52.14 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. A device driver is **software** (system software); the device it controls is hardware (TRAP-10).
2. The OS is **system software**. MS Word is application software; Disk Cleanup is utility software.
3. While running, the OS is held in **primary memory (RAM)**, not executed from secondary storage (PYQ-22).
4. DMA moves data between a **device and main memory** and **bypasses the CPU**; it is not “faster CPU copying”.
5. **RTOS** = Real-Time Operating System, chosen for fixed deadlines.
6. **DOS** = Disk Operating System; Linux distributions include Ubuntu, Fedora, and Debian. **winword** opens MS Word.

Exercises

1. Classify each as system, application, or utility software: Windows, MS Word, Disk Defragmenter, a device driver, Disk Cleanup.
2. Explain why a device driver is described as software even though it controls hardware.
3. Describe a DMA transfer: name the two locations between which data moves and state the role of the CPU.
4. State the full form of RTOS and give one reason an RTOS is used.

5. Which of these are operating systems, and which are Linux distributions: Ubuntu, Windows, Debian, MS Access, Fedora?
6. What does typing `winword` in the Windows Run box do, and how does it differ from `wordpad`?

Solutions

1. System software: Windows and the device driver. Application software: MS Word. Utility software: Disk Defragmenter and Disk Cleanup.
2. The driver is a stored set of instructions that the OS loads and runs to talk to the device. It has no physical body of its own; the hardware is the device. A program that extends OS control of hardware is therefore software, and specifically system software.
3. Data moves directly between the input/output device and main memory. A DMA controller performs the transfer; the CPU starts it and is interrupted when it finishes, but it does not handle each byte.
4. RTOS = Real-Time Operating System. It is used where tasks must complete within fixed time deadlines, as in embedded and control systems.
5. Operating systems: Ubuntu (also a Linux distribution), Windows, Debian (also a Linux distribution), Fedora (also a Linux distribution). MS Access is not an operating system; it is a database application.
6. Typing `winword` opens MS Word. Typing `wordpad` opens WordPad, a different text editor. The two commands open different programs and must not be confused.